```
#include <stdio.h>
#include <ctype.h>

int stack[100], top = -1, size;

void pushval(int x)
{
    if (top < size - 1)
    {
        stack[++top] = x;
    }
}

int popval()
{
    return stack[top--];
}

int calc(int op1, int op2, char op)
{
    int ans, i;
    switch (op)
    {
        case '+':
            ans = op1 + op2;
            break;

        case '-':
            ans = op1 - op2;
            break;

        case '*':
            ans = op1 * op2;
            break;

        case '/':
            ans = op1 / op2;
            break;

        case '^':
            ans = 1;
            for (i = 0; i < op2; i++)
            {
                ans = ans * op1;
            }
            break;

        default:
            ans = 0;
    }
```

```
        else
        {
            op2 = popval();
            op1 = popval();
            ans = calc(op1, op2, post[i]);
            pushval(ans);
        }
    }

    if (top == 0)
    {
        printf("Evaluation is: %d\n", stack[top]);
    }
    else
    {
        printf("Invalid postfix expression\n");
    }
}

int main()
{
    char post[100];

    printf("Enter stack size: ");
    scanf("%d", &size);

    if (size < 1 || size > 100)
    {
        printf("Invalid stack size\n");
        return 0;
    }

    printf("Enter postfix expression: ");
    scanf("%s", post);

    eval(post);

    return 0;
}
```